

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
CHANGA**

M.Sc. (BC/BT/MI) Sem - I Examination (Backlog), 2011-12

BC/BT/MI - 703 Bioinstrumentation and Biostatistics

Date: 18.04.2012, Wednesday

Time: 10.00 a.m to 1.00 p.m

Maximum Marks:70

Instructions:

1. Provide all necessary information as required in the answer sheet.
2. Attempt all questions.
3. Questions in **Part A** should be answered in the question paper itself. Please **do not write** your candidate ID number on the question paper for Part A.
4. Write answers for Section I and Section II of **Part B** in separate answer sheets.
5. Use of cell phones is strictly prohibited.
6. Use of non-programmable calculators may be allowed if needed.

PART A

Total marks: 20

1. Which of the following is also called quantitative ELISA?
(a) indirect ELISA (b) sandwich ELISA (c) competitive ELISA (d) all of the above
2. Living, unstained cells and organisms can be observed best using
a) fluorescent microscopy b) TEM c) phase contrast microscopy d) SEM
3. Transmission electron microscopy is best for high magnification viewing of
a) internal structure of fixed cells. b) internal structure of live, motile cells.
c) surface structure of fixed cells. d) surface membranes of live, motile cells.
4. A highly sensitive technique for measuring antigen or antibody is
a) RIA b) Ouchterlony method c) Agglutination d) All of the above
5. Which of the following radiation source is used in Visible spectroscopy?
(a) Atomisers (b) Deuterium lamp (c) Hollow cathode lamp (d) Tungsten lamp
6. Mass spectrometers separate isotopes of different elements based on their:
a) mass b) electric charge c) mass/charge d) none of these
7. Which of the following radiation source is used in Atomic absorption spectroscopy?
a) Atomisers b) Deuterium lamp c) Hollow cathode lamp d) Xenon discharge lamp
8. Which of the following electromagnetic waves have the highest Energy?
a) Ultraviolet light waves b) X-rays c) Microwaves d) Gamma rays
9. Sorenson's pH scale ranges from
a) 1 to 14 b) 0 to 14 c) 3 to 11 d) 0.1 to 14
10. Relative centrifugal force depends upon
a) RPM b) Radius of rotation c) Both (a) and (b) d) Non of the above
11. The molecular weight of macromolecules can be determined by
a) Ultracentrifugation b) Gel filtration chromatography c) SDS-PAGE
d) All of the above

P.T.O

12. The principle behind the paper chromatography is
a) Adsorption b) Affinity c) Partition coefficient d) Charge
13. Who first developed the DNA sequencing approach using dideoxynucleoside triphosphates in DNA synthesis?
a) Pauling b) Watson c) Sanger d) Edman's
14. Small solid supports onto which are spotted hundreds of thousands of tiny drops of DNA that can be used to screen gene expression.
a) DNA microarrays b) cloning library c) Southern blot d) western blot
15. What key feature of Taq polymerase allows PCR to be conveniently performed?
a) Taq polymerase does not require primers.
b) Taq polymerase does not require a template.
c) Taq polymerase is not damaged by heating.
d) Taq polymerase can work at very low temperatures.
16. What type of probe would you use for a Western blot experiment?
a) A known DNA sequence. b) An RNA molecule. c) A purified protein.
d) An antibody.
17. A measure of central tendency of a set of data is the _____. (Central Value, Positional value)
18. Percentiles divide the total frequency into _____ equal parts.(10, 100)
19. Any measurable characteristic of the universe is called a _____. (Statistic, Parameter)
20. The median is also known as the 50th percentile of a frequency distribution.

(True) (False)

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Part B

Total Marks: 50

SECTION I

- Q1 A Give the principle and application of Phase contrast Microscopy [05]
OR
A Discuss briefly about FISH [05]
B Brief note on radio-immunoassay [05]
OR
B Give the principle and application of Transmission Electron Microscopy [05]
- Q2 A What is the function of each of the following components of UV/Visible spectrophotometer? Radiation source, Photomultiplier tube, prism, exit slit [04]
OR
A Sketch diagram of a atomic absorption spectrophotometer [04]
B Explain the Beer-Lambert law and explore it's significance [03]
OR
B Explain the principle underlying NMR spectroscopy [03]
C Name the component of mass spectrophotometer. What is the function of each? [03]

SECTION II

- Q3 A Give the principle and application of affinity chromatography [05]
OR
A Write the applications of ion exchange chromatography [05]
B Discuss the types of Matrices used in column chromatography [05]
OR
B Discuss the principle and application of SDS-PAGE [05]
- Q4 A Write short note on: (**Any Two**) [3x2 = 6]
Isolation and purification of Plasmid DNA
RFLP
Microarrays
B Briefly discuss PCR [04]

P.T.O

Q5

Attempt (Any Two)

[10]

A Calculate the mean deviation from mean for the following data:

Size	2	4	6	8	10	12	14	16
Frequency	2	2	4	5	3	2	1	1

B Seven measurements of the pH of a buffer solution gave the following results:

5.12 5.20 5.15 5.17 5.16 5.19 5.15

Compute

(a) The mean pH of solution

(b) The standard deviation pH solution

C Group the following 40 measurements of interorbital width of a sample of domestic pigeons into a frequency distribution and draw its histogram. Measurements are in millimeters.

12.2	12.9	11.8	11.9	11.6	11.1	12.3	12.2	11.8	11.8
10.7	11.5	11.3	11.2	11.6	11.9	13.3	11.2	10.5	11.1
12.1	11.9	10.4	10.7	10.8	11.0	11.9	10.2	10.9	11.6
10.8	11.6	10.4	10.7	12.0	12.4	11.7	11.8	11.3	11.1